QuadChain: Proof-Carrying Context Compression for Multiagent AI

white paper

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Abstract

QuadChain turns context compression from a lossy summarization trick into a verifiable systems primitive for multiagent AI. Instead of merely shortening prompts, QuadChain produces proof-carrying compressed memory: low-signal context is deleted, exact downstream evidence is protected, role-specific packets are routed to agents, and every handoff carries hashes, evidence obligations, omission ranges, and registry commitments that receivers can verify before trust. On the current coding-agent benchmark, QuadChain compresses 2,250 tokens to 1,383, saves 867 tokens, preserves 41/41 evidence and 38/38 answer concepts, and keeps answer-readiness at 1.000. Across the actual multi-magnitude scale ladder, QuadChain measures 24 role prompts, preserves 144/144 audited payload evidence checks, and saves 1,253,400 workflow tokens. The frontier benchmark harness adds 2,880 paired rows across 20 cases, 12 methods, four budgets, and three seeds.

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1 Executive Summary

The key claim is not that every task can be compressed by a universal percentage. The claim is narrower and stronger: for coding-agent context, exact downstream evidence can be protected, measured, routed, and audited. For multiagent workflows, the win compounds. The measured 4-agent routing eval drops a monolithic 9,000-token workflow to 2,283 routed tokens, saving 6,717 tokens (74.63%) while preserving 41/41 evidence. Unsafe handoff or route mutations are rejected 4/4 times. The upgraded methodology separately audits payload evidence, certificate evidence, and final prompt evidence so injected obligations cannot count as preserved payload.

2 Architecture

QuadChain has four chains:

1. **Source chain**: input hashes, packet identifiers, and source ranges.
2. **Compression chain**: compressed output hashes, token deltas, and omission ranges.
3. **Proof chain**: required evidence, answer concepts, answer-readiness, and verifier results.
4. **Anchor chain**: Merkle roots and registry receipts that make handoffs tamper-evident while keeping raw context off-chain.

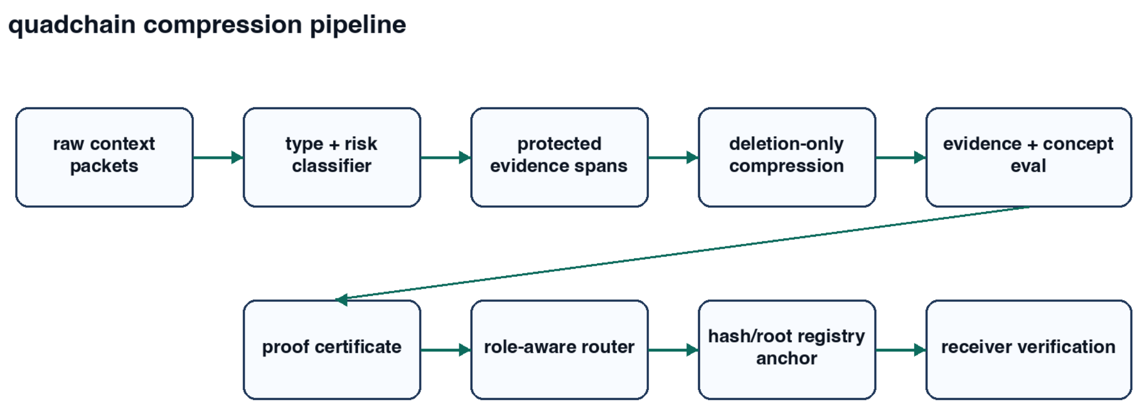

Figure 1: QuadChain compression pipeline.

3 Why Multiagent Compression Is Different

Single-call compression saves prompt tokens once. Multiagent compression saves workflow tokens repeatedly because each downstream agent would otherwise receive duplicated context. QuadChain routes only the compressed packets or proof summaries needed by each role.

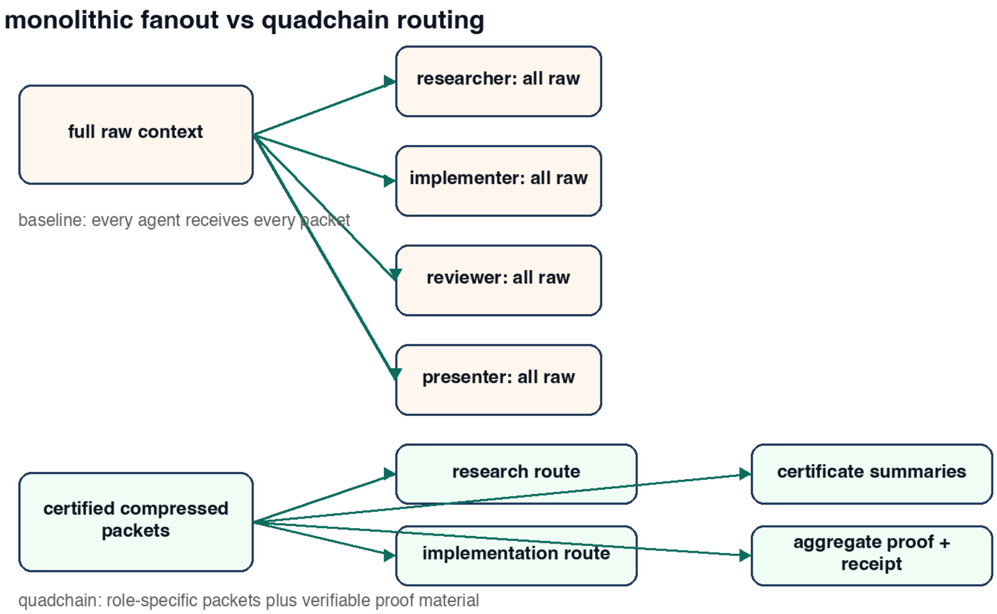

Figure 2: Monolithic fanout versus QuadChain role routing.

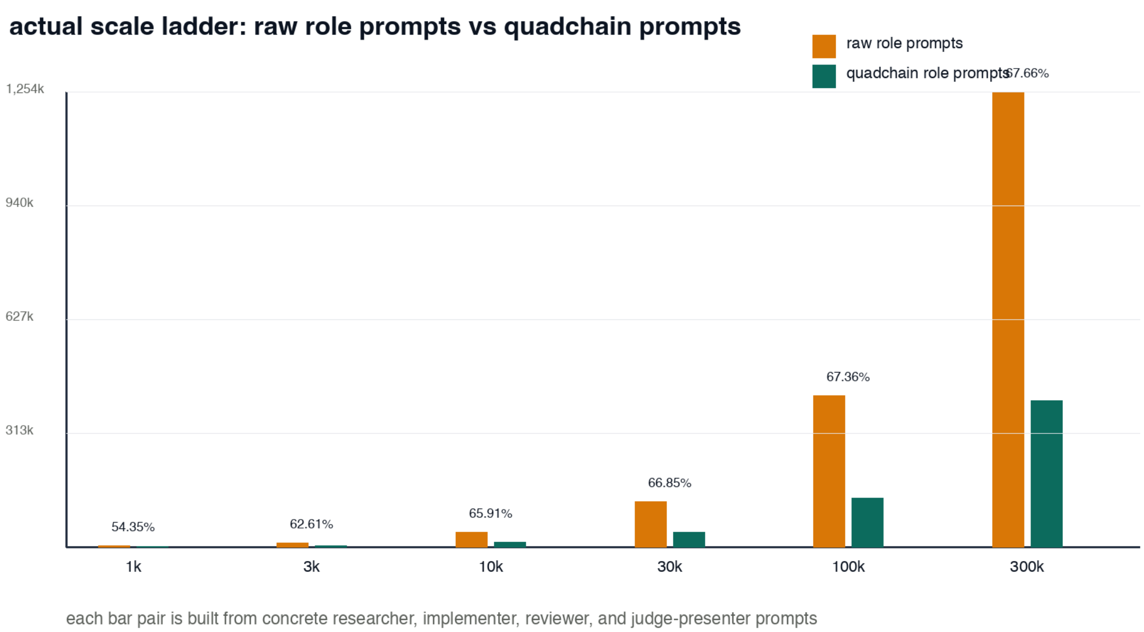

Figure 3: Actual multi-magnitude role-prompt scale ladder.

Table 1: Headline quantitative results.

Eval	Raw	Quad	Saved	Reduction	Evidence	Concepts
Single context	2,250	1,383	867	38.53%	41/41	38/38
4-agent workflow	9,000	2,283	6,717	74.63%	41/41	38/38
Measured large context	115,038	74,813	40,225	34.97%	12/12	n/a
Scale ladder actual prompts	1,858,850	605,450	1,253,400	67.43%	36/36	n/a
Frontier benchmark harness	n/a	budget matched	n/a	57.85% mean	144/144	score 0.6350

4 Measured Results

5 Frontier Benchmark Methodology

The frontier harness emits one row per dataset adapter, item id, method, budget, seed, and scorer. The current run contains 2,880 paired rows across 20 cases, 12 methods, four budgets, and three seeds. It includes local fixtures plus needle-style retrieval, ruler-style multihop, and longbench-style QA adapters. These are public-benchmark-inspired local slices, not claims that the full external public datasets were downloaded and run.

Table 2: Frontier methodology checks.

Surface	Result	Purpose
Paired rows	2,880	Same item, budget, method, seed schema.
Methods	12	Raw, QuadChain, truncation, retrieval, summary, and proxy learned baselin
Role payload audit	144/144	Required evidence text cannot satisfy itself.
Receiver task success	75.0%	Receiving agents must complete role-specific tasks.
Receiver workflow savings	74.22%	Multiagent routing saves context repeatedly.

6 Verification Model

The receiver should never trust a compressed summary just because another agent produced it. It verifies certificate hashes, Merkle roots, verifier versions, registry commitments, evidence obligations, answer concepts, and route obligations before acting on the handoff.

7 Threat Model

8 Implementation Status

- Deterministic local compressor, CLI, batch runner, and budget router.
- Proof certificate generator with omission ranges.
- Multiagent handoff verifier with adversarial cases.
- QuadChain eval report and machine-readable JSON.
- Frontier benchmark harness with 2,880 paired rows, role evidence audit, receiver task eval, and bootstrap confidence intervals.

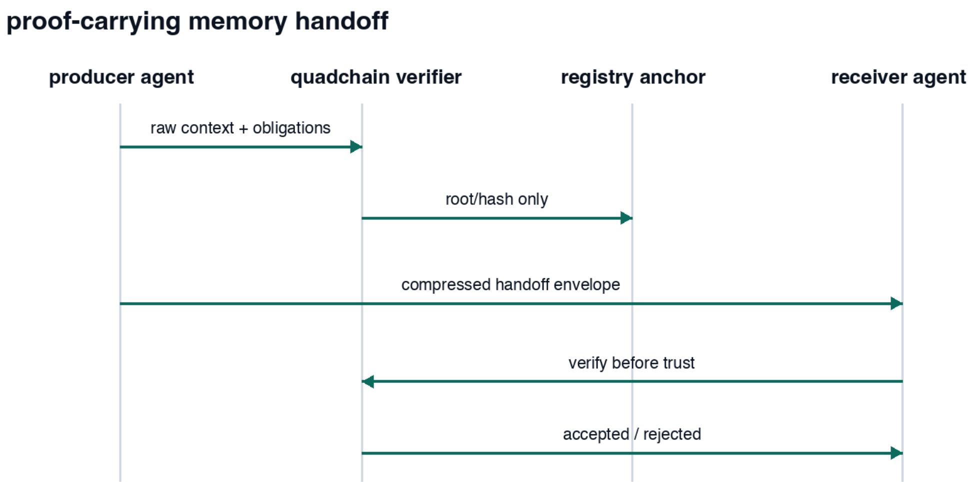

Figure 4: Proof-carrying memory handoff.

same-budget baseline failure surface

system	evidence	answer concepts	failure rows
quadchain	**41/41**	**38/38**	**0 failure rows**
best naive	41/41	36/38	20 failure rows

- headline: quadchain keeps exact evidence and answer concepts under same-budget pressure
- baselines include head, middle, tail, even-line, and deterministic random-line deletion
- failure rows mean at least one required evidence item or answer concept was lost

Figure 5: Same-budget baseline failure surface.

Table 3: Threats and controls.

Threat	Failure Mode	QuadChain Response
Blind truncation	Drops middle or tail facts	Same-budget baseline eval exposes missing evidence or concepts.
Unsafe compression	Deletes exact ids, paths, errors, or dates	Protected spans and answer-readiness gate.
Malicious producer	Alters compressed memory after certification	Certificate hash and Merkle root mismatch.
Stale receipt	Reuses old registry proof	Handoff hash mismatch.
Over-sharing on-chain	Leaks raw context or evidence strings	Only hashes, roots, and validation metadata are anchored.
Invalid route	Omits role-critical packets	Route validation rejects missing obligations.

- Comprehensive test suite with 46/46 checks passing.
- Submission bundle with scripts, fixtures, reports, PDF papers, and verification command.

9 Limits

QuadChain is not a claim that this exact local compressor is state of the art. LLMLingua, LongLLMLingua, LLMLingua-2, and Selective Context explore stronger model-based compression families. The contribution here is the proof-carrying control plane: evidence obligations, role routing, tamper-evident memory handoffs, and deterministic judge-ready evals for coding-agent context.

References

[1] Jiang et al. *LLMLingua: Compressing Prompts for Accelerated Inference of Large Language Models.* arXiv:2310.05736. https://arxiv.org/abs/2310.05736

[2] Jiang et al. *LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression.* arXiv:2310.06839. https://arxiv.org/abs/2310.06839

[3] Pan et al. *LLMLingua-2: Data Distillation for Efficient and Faithful Task-Agnostic Prompt Compression.* arXiv:2403.12968. https://arxiv.org/abs/2403.12968

[4] Li. *Selective Context for LLMs.* https://github.com/liyucheng09/Selective_Context

[5] Zhou et al. *Prompt Compression for Large Language Models: A Survey.* NAACL 2025. https://aclanthology.org/2025.naacl-long.368.pdf

[6] Ethereum Improvement Proposals. *ERC-8004: Trustless Agents.* https://eips.ethereum.org/EIPS/eip-8004

[7] Ethereum Improvement Proposals. *ERC-8126: AI Agent Verification.* https://eips.ethereum.org/EIPS/eip-8126

[8] Eco. *TEEs for AI Agents: Verifiable Compute.* https://eco.com/support/en/articles/14796365-tees-for-ai-agents-verifiable-compute